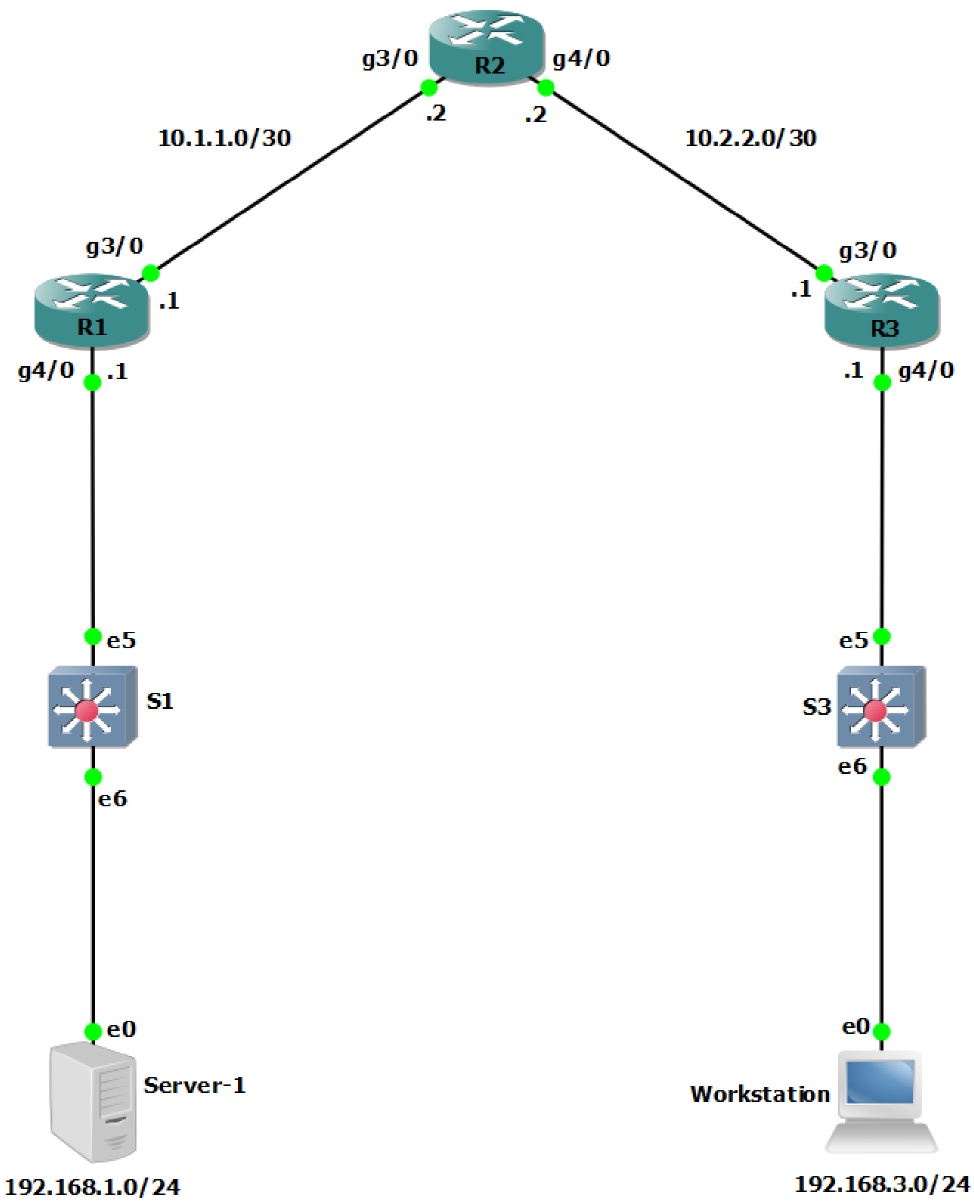
Lab - Configure Local AAA Authentication (DAYUTH THY)

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G3/0	10.1.1.1	255.255.255.252	N/A	N/A
	G4/0	192.168.1.1	255.255.255.0	N/A	S1 E5

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R2	G3/0	10.1.1.2	255.255.255.252	N/A	N/A
	G4/0	10.2.2.2	255.255.255.252	N/A	N/A
R3	G3/0	10.2.2.1	255.255.255.252	N/A	N/A
	G4/0	192.168.3.1	255.255.255.0	N/A	S3 E5
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 E6
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 E6

Objectives

Part 1: Configure Basic Device Settings

- Configure basic settings such as host name, interface IP addresses, and access passwords.
- Configure static routing.

Part 2: Configure Local Authentication for Console Access

- Configure a local database user and local access for the console line.
- Test the configuration.

Part 3: Configure Local Authentication for Remote Access

- Configure domain name
- Configure encryption key
- Enable SSH on vty

Part 4: Configure Local Authentication Using AAA on R3

- Configure the local user database using Cisco IOS.
- Configure AAA local authentication using Cisco IOS.
- Test the configuration.

Part 5: Observe AAA Authentication Using Cisco IOS Debug

Background / Scenario

The most basic form of router access security is to create passwords for the console, vty, and aux lines. A user is prompted for only a password when accessing the router. Configuring a privileged EXEC mode enable secret password further improves security, but still only a basic password is required for each mode of access.

In addition to basic passwords, specific usernames or accounts with varying privilege levels can be defined in the local router database that can apply to the router as a whole. When the console, vty, or aux lines are configured to refer to this local database, the user is prompted for a username and a password when using any of these lines to access the router.

Additional control over the login process can be achieved using authentication, authorization, and accounting (AAA). For basic authentication, AAA can be configured to access the local database for user logins, and fallback procedures can also be defined. However, this approach is not very scalable because it must be configured on every router. When a user attempts to log in, the router references the external server database to verify that the user is logging in with a valid username and password.

In this lab, you build a multi-router network and configure the routers and hosts. You will then use CLI commands to configure routers with basic local authentication by means of AAA.

Note: The routers used with hands-on labs are Cisco 4221 with Cisco IOS XE Release 16.9.6 (universalk9 image). The switches used in the labs are Cisco Catalyst 2960+ with Cisco IOS Release 15.2(7) (lanbasek9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs. Refer to the Router Interface Summary Table at the end of the lab for the correct interface identifiers.

Note: Before you begin, ensure that the routers and the switches have been erased and have no startup configurations.

Required Resources

- 3 Routers (Cisco 4221 with Cisco XE Release 16.9.6 universal image or comparable with a Security Technology Package license)
- 2 Switches (Cisco 2960+ with Cisco IOS Release 15.2(7) lanbasek9 image or comparable)
- 2 PCs (Windows OS with a terminal emulation program, such as PuTTY or Tera Term installed)
- Console cables to configure Cisco networking devices
- Ethernet cables as shown in the topology

Instructions

Part 1: Configure Basic Device Settings

In this part, set up the network topology and configure basic settings, such as interface IP addresses.

Step 1: Cable the network.

Attach the devices, as shown in the topology diagram, and cable as necessary.

Step 2: Configure basic settings for each router.

a. Console into the router and enable privileged EXEC mode.

```
Router> enable
Router# configure terminal
```

Configure host names as shown in the topology.

```
R1(config)# hostname R1
```

Configure interface IP addresses as shown in the IP Addressing Table.

```
R1(config)# interface g0/0/0
R1(config-if)# ip address 10.1.1.1 255.255.255.0
R1(config-if)# no shutdown

R1(config)# interface g0/0/1
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
```

```
interface GigabitEthernet3/0
 ip address 10.1.1.1 255.255.255.252
 negotiation auto
!
interface GigabitEthernet4/0
 ip address 192.168.1.1 255.255.255.0
 negotiation auto
!
interface GigabitEthernet5/0
 no ip address
 shutdown
 negotiation auto
!
router ospf 1
 network 10.1.1.0 0.0.0.3 area 0
 network 192.168.1.0 0.0.0.255 area 0
!
```

b. To prevent the router from attempting to translate incorrectly entered commands as though they were host names, disable DNS lookup. R1 is shown here as an example.

```
R1(config)# no ip domain-lookup
```

```
no ip domain lookup
```

Step 3: Configure OSPF routing on the routers.

a. Use the **router ospf** command in global configuration mode to enable OSPF on R1.

```
R1(config)# router ospf 1
```

b. Configure the **network** statements for the networks on R1. Use an area ID of 0.

```
R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
R1(config-router)# network 10.1.1.0 0.0.0.3 area 0
```

```
router ospf 1
 passive-interface GigabitEthernet4/0
 network 10.1.1.0 0.0.0.3 area 0
 network 192.168.1.0 0.0.0.255 area 0
```

c. Configure OSPF on R2 and R3.

```
R2(config)# router ospf 1
R2(config-router)# network 10.1.1.0 0.0.0.3 area 0
R2(config-router)# network 10.2.2.0 0.0.0.3 area 0
```

```
router ospf 1
 network 10.1.1.0 0.0.0.3 area 0
 network 10.2.2.0 0.0.0.3 area 0
```

```
R3(config)# router ospf 1
R3(config-router)# network 10.2.2.0 0.0.0.3 area 0
```

```
R3(config-router)# network 192.168.3.0 0.0.0.255 area 0
```

```
router ospf 1
 passive-interface GigabitEthernet4/0
 network 10.2.2.0 0.0.0.3 area 0
 network 192.168.3.0 0.0.0.255 area 0
```

d. Issue the **passive-interface** command to change the G0/0/1 interface on R1 and R3 to passive.

```
R1(config)# router ospf 1
R1(config-router)# passive-interface g0/0/1
```

```
R1(config)#router ospf 1
R1(config-router)#pas
R1(config-router)#passive-interface g/4
                                      ^
% Invalid input detected at '^' marker.

R1(config-router)#passive-interface g4/0
```

```
R3(config)# router ospf 1
R3(config-router)# passive-interface g0/0/1
```

```
R3(config)#router ospf 1
R3(config-router)#pass
R3(config-router)#passive-interface g4/0
```

Step 4: Verify OSPF neighbors and routing information.

a. Issue the **show ip ospf neighbor** command to verify that each router lists the other routers in the network as neighbors.

```
R1# show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
10.2.2.2          1   FULL/BDR        00:00:37    10.1.1.2        GigabitEthernet0/0/0
```

```
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
10.2.2.2          1   FULL/BDR        00:00:39    10.1.1.2        GigabitEthernet3/0
```

b. Issue the **show ip route** command to verify that all networks display in the routing table on all routers.

```
R1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C        10.1.1.0/30 is directly connected, GigabitEthernet0/0/0
L        10.1.1.1/32 is directly connected, GigabitEthernet0/0/0
```

```
O        10.2.2.0/30 [110/2] via 10.1.1.2, 00:01:11, GigabitEthernet0/0/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, GigabitEthernet0/0/1
L        192.168.1.1/32 is directly connected, GigabitEthernet0/0/1
```

- 192.168.3.0/24 [110/3] via 10.1.1.2, 00:01:07, GigabitEthernet0/0/0

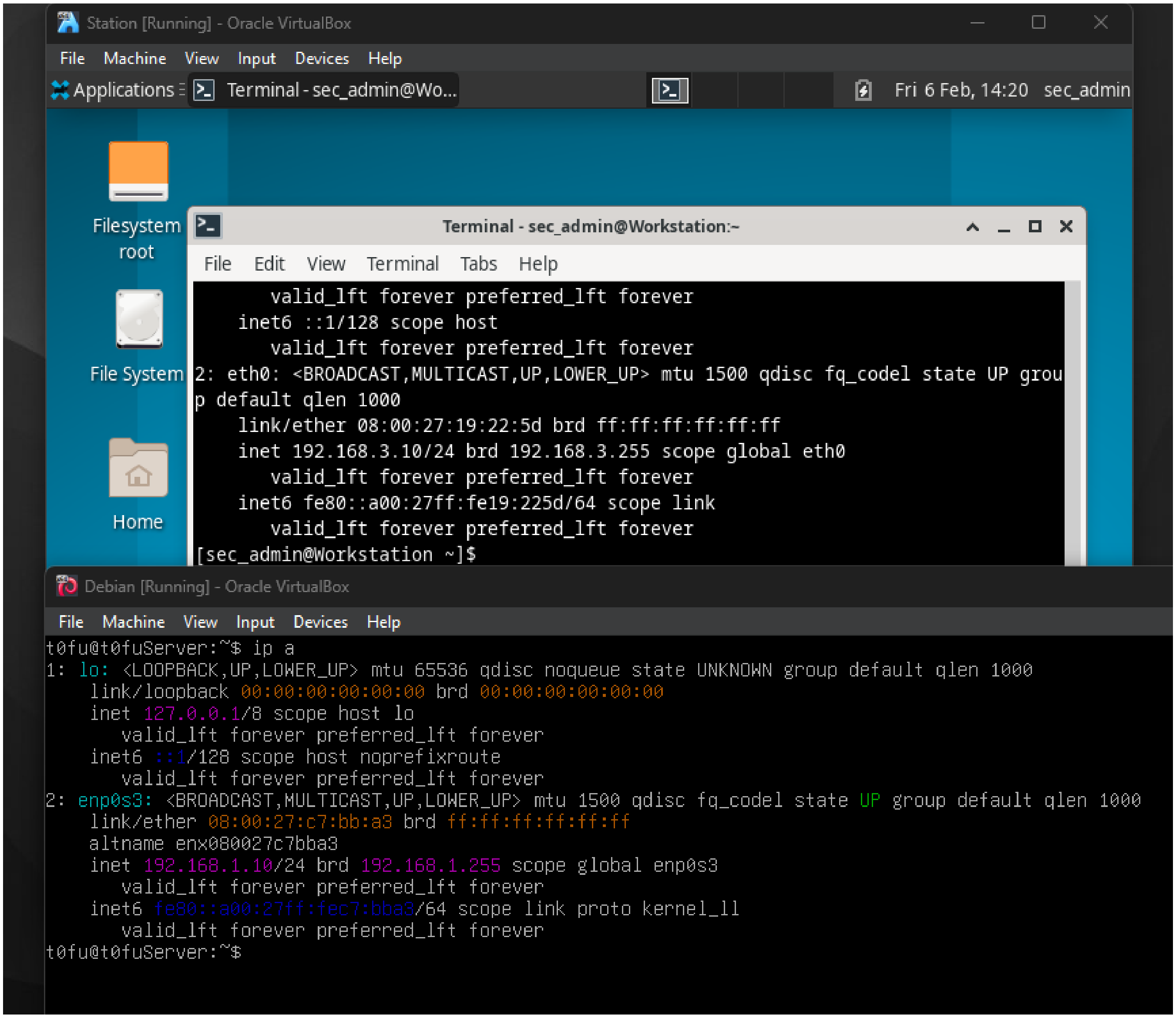
```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C        10.1.1.0/30 is directly connected, GigabitEthernet3/0
L        10.1.1.1/32 is directly connected, GigabitEthernet3/0
O        10.2.2.0/30 [110/2] via 10.1.1.2, 01:56:14, GigabitEthernet3/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, GigabitEthernet4/0
L        192.168.1.1/32 is directly connected, GigabitEthernet4/0
O     192.168.3.0/24 [110/3] via 10.1.1.2, 01:43:25, GigabitEthernet3/0
```

Step 5: Configure PC host IP settings.

Configure a static IP address, subnet mask, and default gateway for PC-A and PC-C as shown in the IP Addressing Table.



Step 6: Verify connectivity between PC-A and PC-C.

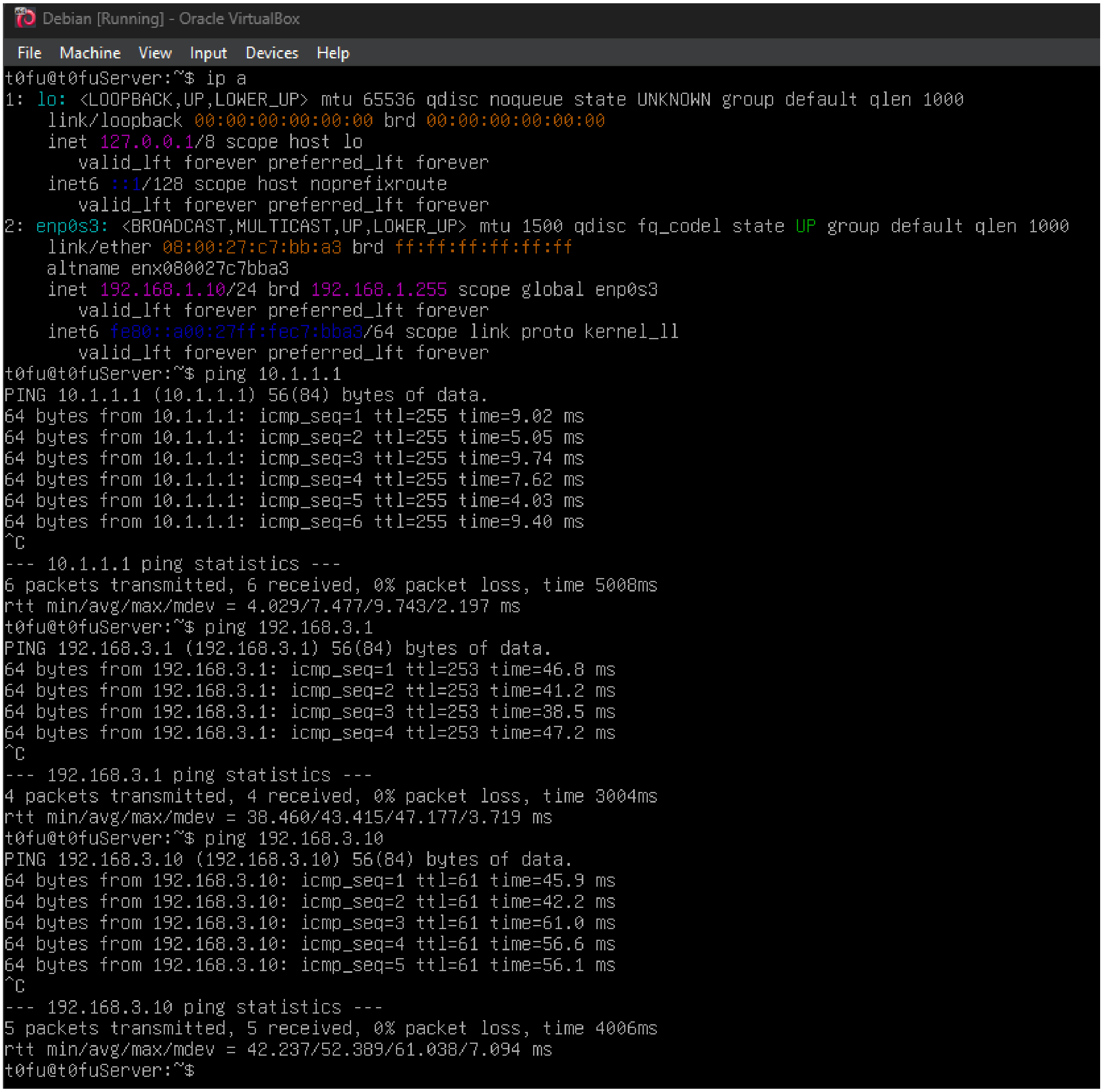
a. Ping from R1 to R3.

   If the pings are not successful, troubleshoot the basic device configurations before continuing.

b. Ping from PC-A, on the R1 LAN, to PC-C, on the R3 LAN.

   If the pings are not successful, troubleshoot the basic device configurations before continuing.

Note: If you can ping from PC-A to PC-C you have demonstrated that OSPF routing is configured and functioning correctly. If you cannot ping but the device interfaces are up and IP addresses are correct, use the **show run, show ip ospf neighbor,** and **show ip route** commands to help identify routing protocol-related problems.



```
Debian [Running] - Oracle VirtualBox
File Machine View Input Devices Help
t0fu@t0fuServer:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:c7:bb:a3 brd ff:ff:ff:ff:ff:ff
    altname enx080027c7bba3
    inet 192.168.1.10/24 brd 192.168.1.255 scope global enp0s3
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fec7:bba3/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
t0fu@t0fuServer:~$ ping 10.1.1.1
PING 10.1.1.1 (10.1.1.1) 56(84) bytes of data.
64 bytes from 10.1.1.1: icmp_seq=1 ttl=255 time=9.02 ms
64 bytes from 10.1.1.1: icmp_seq=2 ttl=255 time=5.05 ms
64 bytes from 10.1.1.1: icmp_seq=3 ttl=255 time=9.74 ms
64 bytes from 10.1.1.1: icmp_seq=4 ttl=255 time=7.62 ms
64 bytes from 10.1.1.1: icmp_seq=5 ttl=255 time=4.03 ms
64 bytes from 10.1.1.1: icmp_seq=6 ttl=255 time=9.40 ms
^C
--- 10.1.1.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5008ms
rtt min/avg/max/mdev = 4.029/7.477/9.743/2.197 ms
t0fu@t0fuServer:~$ ping 192.168.3.1
PING 192.168.3.1 (192.168.3.1) 56(84) bytes of data.
64 bytes from 192.168.3.1: icmp_seq=1 ttl=253 time=46.8 ms
64 bytes from 192.168.3.1: icmp_seq=2 ttl=253 time=41.2 ms
64 bytes from 192.168.3.1: icmp_seq=3 ttl=253 time=38.5 ms
64 bytes from 192.168.3.1: icmp_seq=4 ttl=253 time=47.2 ms
^C
--- 192.168.3.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 38.460/43.415/47.177/3.719 ms
t0fu@t0fuServer:~$ ping 192.168.3.10
PING 192.168.3.10 (192.168.3.10) 56(84) bytes of data.
64 bytes from 192.168.3.10: icmp_seq=1 ttl=61 time=45.9 ms
64 bytes from 192.168.3.10: icmp_seq=2 ttl=61 time=42.2 ms
64 bytes from 192.168.3.10: icmp_seq=3 ttl=61 time=61.0 ms
64 bytes from 192.168.3.10: icmp_seq=4 ttl=61 time=56.6 ms
64 bytes from 192.168.3.10: icmp_seq=5 ttl=61 time=56.1 ms
^C
--- 192.168.3.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 42.237/52.389/61.038/7.094 ms
t0fu@t0fuServer:~$
```

Step 7: Configure the basic console, auxiliary port, and vty lines.

a. Configure a console password and enable login for router R1. For additional security, the **exec-timeout** command causes the line to log out after **5** minutes of inactivity. The **logging synchronous** command prevents console messages from interrupting command entry.

   Note: To avoid repetitive logins during this lab, the exec timeout can be set to 0 0, which prevents it from expiring. However, this is not considered a good security practice.

```
R1(config)# line console 0
R1(config-line)# password ciscoconpass
R1(config-line)# exec-timeout 5 0
```

```
R1(config-line)# login
R1(config-line)# logging synchronous
```

b. Configure a password for the aux port for router R1.

```
R1(config)# line aux 0
R1(config-line)# password ciscoauxpass
R1(config-line)# exec-timeout 5 0
R1(config-line)# login
```

c. Configure the password on the vty lines for router R1.

```
R1(config)# line vty 0 4
R1(config-line)# password ciscovtypass
R1(config-line)# exec-timeout 5 0
R1(config-line)# login
```

d. Encrypt the console, aux, and vty passwords.

```
R1(config)# service password-encryption
```

e. Issue the **show run | section line** command.

Can you read the console, aux, and vty passwords? Explain.

```
line con 0
 exec-timeout 5 0
 privilege level 15
 password 7 070C285F4D061A0A19020A1F17
 logging synchronous
 login
 stopbits 1
line aux 0
 exec-timeout 5 0
 privilege level 15
 password 7 1511021F07252A313023343100
 logging synchronous
 login
 stopbits 1
line vty 0 4
 exec-timeout 5 0
 password 7 094F471A1A0A0106121C053938
 login
 transport input all
!
```

No, it's encrypted

Step 8: Configure a login warning banner on routers R1 and R3.

a. Configure a warning to unauthorized users using a message-of-the-day (MOTD) banner with the **banner motd** command. When a user connects to the router, the MOTD banner appears before the login prompt. In this example, the dollar sign ($) is used to start and end the message.

```
R1(config)# banner motd $Unauthorized access strictly prohibited!$
R1(config)# exit
```

b. Exit privileged EXEC mode by using the **disable** or **exit** command and press **Enter** to get started.

If the banner does not appear correctly, re-create it using the **banner motd** command.

```
R1 con0 is now available



Press RETURN to get started.

Unauthorized access strictly prohibited!

User Access Verification

Password:
R1#
```

Step 9: Save the basic configurations on all routers.

Save the running configuration to the startup configuration from the privileged EXEC prompt.

```
R1# copy running-config startup-config
```

```
R1#copy running-config st
R1#copy running-config startup-config
Destination filename [startup-config]?
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R1#wr
Building configuration...
[OK]
R1#
```

Part 2: Configure Local Authentication for Console Access

In this part of lab, you configure a local username and password and change the access for the console, aux, and vty lines to reference the router's local database for valid usernames and passwords. Perform all steps on R1 and R3. The procedure for R1 is shown here.

Step 1: Configure the local user database.

a. Create a local user account using the type 8 (PDKDF2) hashing algorithm to encrypt the password.

```
R1(config)# username user01 algorithm-type sha256 secret user01pass
```

```
R1(config)#username user01 algorithm-type sha256 secret user01pass
R1(config)#exit
R1#
*Feb  7 11:26:14.155: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

b. Exit global configuration mode and display the running configuration.

Can you read the user's password?

No, a secret password is encrypted

Step 2: Configure local authentication for the console line and login.

a. Set the console line to use the locally defined login usernames and passwords.

```
R1(config)# line console 0
R1(config-line)# login local
```

b. Exit to the initial router screen that displays:

```
R1 con0 is now available.

Press RETURN to get started.
```

c. Log in using the **user01** account and password previously defined.

```
Unauthorized access strictly prohibited!

User Access Verification

Username: user01
Password:
R1>
```

What is the difference between logging in at the console now and previously?

Need username and password

d. After logging in, issue the **show run** command.

Were you able to issue the command? Explain

No. It requires privileged EXEC level

e. Enter privileged EXEC mode using the **enable** command.

Were you prompted for a password? Explain.

Yes. The new users created will still be required to enter the enable secret password to enter privileged EXEC mode

```
!
!
!
username user01 secret 8 $8$WGpM7owJb4S9bj$TxyyhTCX2Lwas1iaRTf5aJn3GWp2zQyi0HZ/p/aFRpM
!
redundancy
```

Part 3: Configure Local Authentication for Remote Access

In this part, you will use SSH for remote access to R1 using local user database.

Step 1: Configure a domain name for the device.

Step 2: Configure the encryption key method.

Step 3: Enable SSH on the vty lines.

a. Enable SSH on the inbound vty lines using the **transport input** command.

b. Change the login method to use the local database for user verification.

c. From PC-A, establish an SSH session with R1.

Were you prompted for a user account? Explain.

Yes, you were prompted for a user account. You used username user01 and password user01pass to establish an SSH session with R1

d. While connected to R1 via SSH, access privileged EXEC mode with the **enable** command.

What password did you use?

cisco12345

e. For added security, set the aux port to use the locally defined login accounts.

```
R1(config)# line aux 0
R1(config-line)# login local
```

Step 4: Save the configuration on R1.

Save the running configuration to the startup configuration from the privileged EXEC prompt.

```
R1# copy running-config startup-config
```

Step 5: Perform steps 1 through 4 on R3 and save the configuration.

Part 4: Configure Local Authentication Using AAA on R3

Step 1: Configure the local user database.

a. Create a local user account with PDKDF2 hashing to encrypt the password.

```
R3(config)# username Admin01 privilege 15 algorithm-type sha256 secret
Admin01pass
```

b. Exit global configuration mode and display the running configuration.

Can you read the user's password?

No, the password is encrypted

Step 2: Enable AAA services.

On R3, enable services with the global configuration **aaa new-model** command. Because you are implementing local authentication, use local authentication as the first method, and no authentication as the secondary method.

If you were using an authentication method with a remote server, such as TACACS+ or RADIUS, you would configure a secondary authentication method for fallback if the server is unreachable. Normally, the secondary method is the local database. In this case, if no usernames are configured in the local database, the router allows all users login access to the device.

```
R3(config)# aaa new-model
```

Step 3: Implement AAA services for console access using the local database.

a. Create the default login authentication list by issuing the **aaa authentication login default** *method1[method2][method3]* command with a method list using the **local** and **none** keywords.

```
R3(config)# aaa authentication login default local-case none
```

Note: If you do not set up a default login authentication list, you could get locked out of the router and be forced to use the password recovery procedure for your specific router.

Note: The **local-case** parameter is used to make usernames case-sensitive.

```
R3(config)#aaa new-mo
R3(config)#aaa new-model
R3(config)#aaa authentication login default local-case none
R3(config)#
```

b. Exit to the initial router screen that displays:

```
R3 con0 is now available


Press RETURN to get started.
```

Log in to the console as **Admin01** with a password of **Admin01pass**. Remember that usernames and passwords are both case-sensitive now.

Were you able to log in? Explain.

Yes. The router verified the account against the local database

Note: If your session with the console port of the router times out, you might have to log in using the default authentication list.

c. Exit to the initial router screen that displays:

d. Attempt to log in to the console as **baduser** with any password.

Were you able to log in? Explain.

Yes. If the username is not found in the local database, the none option on the command aaa authentication login default local none requires no authentication

If no user accounts are configured in the local database, which users are permitted to access the device?

Any users can access the device. It does not matter whether the username exists in the local database or if the password is correct.

Step 4: Create an AAA authentication profile for SSH using the local database.

a. Create a unique authentication list for SSH access to the router. This does not have the fallback of no authentication, so if there are no usernames in the local database, SSH access is disabled. To create an authentication profile that is not the default, specify a list name of SSH_LINES and apply it to the vty lines.


```
R3(config)# aaa authentication login SSH_LINES local
R3(config)# line vty 0 4
R3(config-line)# login authentication SSH_LINES
```

b. Verify that this authentication profile is used by opening an SSH session from PC-C to R3. Log in as **Admin01** with a password of **Admin01pass**.

   Were you able to login? Explain.

   Yes, it's in the database

c. Exit the SSH session.

d. Attempt to log in as **baduser** with any password.

   Were you able to login? Explain.

   No. If the username is not found in the local database, there is no fallback method specified in the authentication list for the vty lines

Part 5: Observe AAA Authentication Using Cisco IOS Debug

In this part, you use the **debug** command to observe successful and unsuccessful authentication attempts.

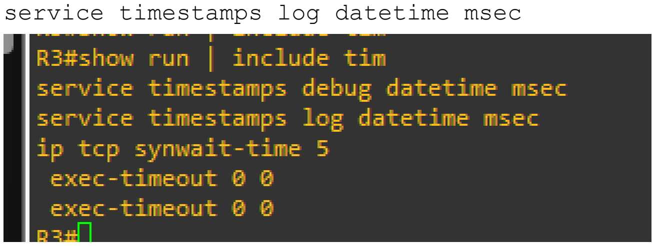
Step 1: Verify that the system clock and debug time stamps are configured correctly.

a. From the R3 user or privileged EXEC mode prompt, use the **show clock** command to determine what the current time is for the router. If the time and date are incorrect, set the time from privileged EXEC mode with the command **clock set HH:MM:SS DD month YYYY.** An example is provided here for R3.

```
R3# clock set 14:15:00 03 February 2021
```

b. Verify that detailed time-stamp information is available for your debug output using the **show run** command. This command displays all lines in the running config that include the text "timestamps".

```
R3# show run | include timestamps
service timestamps debug datetime msec
service timestamps log datetime msec
```



c. If the **service timestamps debug** command is not present, enter it in global config mode.

```
R3(config)# service timestamps debug datetime msec
R3(config)# exit
```

d. Save the running configuration to the startup configuration from the privileged EXEC prompt.

```
R3# copy running-config startup-config
```

Step 2: Use debug to verify user access.

a. Activate debugging for AAA authentication.

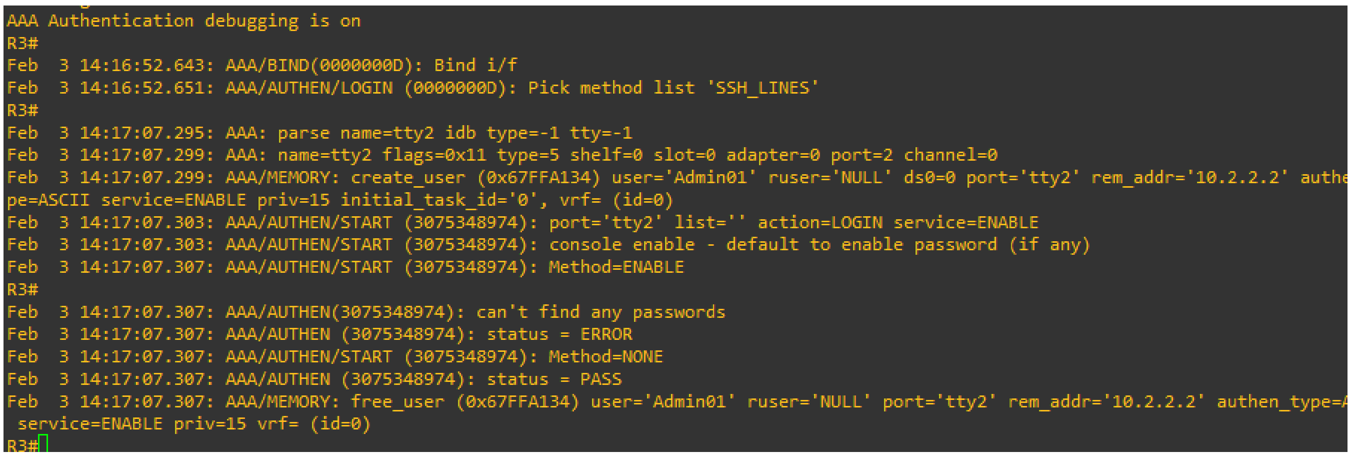
```
R3# debug aaa authentication
AAA Authentication debugging is on
```

b. Start an SSH session from R2 to R3. Log in with username **Admin01** and password **Admin01pass**.

```
R2# ssh -l Admin01 10.2.2.1
```

c. Navigate back R3. Observe the AAA authentication events in the console session window. Debug messages similar to the following should be displayed.

```
R3#
Feb  3 14:15:57.653: AAA/BIND(00000FB5): Bind i/f
Feb  3 14:15:57.653: AAA/AUTHEN/LOGIN (00000FB5): Pick method list 'SSH_LINES'
R3#
Feb  3 14:16:01.966: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: Admin01]
[Source: 10.2.2.2] [localport: 22] at 14:16:01 UTC Wed Feb 3 2021
```



```
AAA Authentication debugging is on
R3#
Feb  3 14:16:52.643: AAA/BIND(0000000D): Bind i/f
Feb  3 14:16:52.651: AAA/AUTHEN/LOGIN (0000000D): Pick method list 'SSH_LINES'
R3#
Feb  3 14:17:07.295: AAA: parse name=tty2 idb type=-1 tty=-1
Feb  3 14:17:07.299: AAA: name=tty2 flags=0x11 type=5 shelf=0 slot=0 adapter=0 port=2 channel=0
Feb  3 14:17:07.299: AAA/MEMORY: create_user (0x67FFA134) user='Admin01' ruser='NULL' ds0=0 port='tty2' rem_addr='10.2.2.2' authe
pe=ASCII service=ENABLE priv=15 initial_task_id='0', vrf= (id=0)
Feb  3 14:17:07.303: AAA/AUTHEN/START (3075348974): port='tty2' list='' action=LOGIN service=ENABLE
Feb  3 14:17:07.303: AAA/AUTHEN/START (3075348974): console enable - default to enable password (if any)
Feb  3 14:17:07.307: AAA/AUTHEN/START (3075348974): Method=ENABLE
R3#
Feb  3 14:17:07.307: AAA/AUTHEN(3075348974): can't find any passwords
Feb  3 14:17:07.307: AAA/AUTHEN (3075348974): status = ERROR
Feb  3 14:17:07.307: AAA/AUTHEN/START (3075348974): Method=NONE
Feb  3 14:17:07.307: AAA/AUTHEN (3075348974): status = PASS
Feb  3 14:17:07.307: AAA/MEMORY: free_user (0x67FFA134) user='Admin01' ruser='NULL' port='tty2' rem_addr='10.2.2.2' authen_type=A
 service=ENABLE priv=15 vrf= (id=0)
R3#
```

d. From the SSH window on R2, enter privileged EXEC mode. Use the enable secret password of **cisco12345**. Debug messages similar to the following should be displayed. In the third entry, note the username (Admin01), virtual port number (tty866), and remote SSH client address (10.2.2.2). Also note that the last status entry is "PASS."

```
Feb  3 14:19:51.146: AAA: parse name=tty866 idb type=-1 tty=-1
Feb  3 14:19:51.146: AAA: name=tty866 flags=0x11 type=5 shelf=0 slot=0 adapter=0
port=866 channel=0
Feb  3 14:19:51.146: AAA/MEMORY: create_user (0x7FD084CE0FF0) user='Admin01'
ruser='NULL' ds0=0 port='tty866' rem_addr='10.2.2.2' authen_type=ASCII service=ENABLE
priv=15 initial_task_id='0', vrf= (id=0)
Feb  3 14:19:51.146: AAA/AUTHEN/START (402765494): port='tty866' list='' action=LOGIN
service=ENABLE
Feb  3 14:19:51.146: AAA/AUTHEN/START (402765494): non-console enable - default to
enable password
Feb  3 14:19:51.147: AAA/AUTHEN/START (402765494): Method=ENABLE
R3#
Feb  3 14:19:51.147: AAA/AUTHEN (402765494): status = GETPASS
R3#
Feb  3 14:19:54.156: AAA/AUTHEN/CONT (402765494): continue_login (user='(undef)')
Feb  3 14:19:54.156: AAA/AUTHEN (402765494): status = GETPASS
Feb  3 14:19:54.156: AAA/AUTHEN/CONT (402765494): Method=ENABLE
Feb  3 14:19:54.259: AAA/AUTHEN (402765494): status = PASS
Feb  3 14:19:54.259: AAA/MEMORY: free_user (0x7FD084CE0FF0) user='NULL' ruser='NULL'
port='tty866' rem_addr='10.2.2.2' authen_type=ASCII service=ENABLE priv=15 vrf= (id=0)
```


e. From the SSH window, exit privileged EXEC mode using the **disable** command. Try to enter privileged EXEC mode again, but use a bad password this time. Observe the debug output on R3, noting that the status is "FAIL" this time.

```
Feb  3 14:24:20.274: AAA: parse name=tty866 idb type=-1 tty=-1
Feb  3 14:24:20.274: AAA: name=tty866 flags=0x11 type=5 shelf=0 slot=0 adapter=0
port=866 channel=0
Feb  3 14:24:20.274: AAA/MEMORY: create_user (0x7FD08991D130) user='Admin01'
ruser='NULL' ds0=0 port='tty866' rem_addr='10.2.2.2' authen_type=ASCII service=ENABLE
priv=15 initial_task_id='0', vrf= (id=0)
Feb  3 14:24:20.274: AAA/AUTHEN/START (1943266075): port='tty866' list='' action=LOGIN
service=ENABLE
Feb  3 14:24:20.274: AAA/AUTHEN/START (1943266075): non-console enable - default to
enable password
Feb  3 14:24:20.274: AAA/AUTHEN/START (1943266075): Method=ENABLE
R3#
Feb  3 14:24:20.275: AAA/AUTHEN (1943266075): status = GETPASS
R3#
Feb  3 14:24:22.276: AAA/AUTHEN/CONT (1943266075): continue_login (user='(undef)')
Feb  3 14:24:22.276: AAA/AUTHEN (1943266075): status = GETPASS
Feb  3 14:24:22.276: AAA/AUTHEN/CONT (1943266075): Method=ENABLE
Feb  3 14:24:22.379: AAA/AUTHEN(1943266075): password incorrect
Feb  3 14:24:22.379: AAA/AUTHEN (1943266075): status = FAIL
Feb  3 14:24:22.379: AAA/MEMORY: free_user (0x7FD08991D130) user='NULL' ruser='NULL'
port='tty866' rem_addr='10.2.2.2' authen_type=ASCII service=ENABLE priv=15 vrf= (id=0)
R3#
```

f. Exit the SSH session to the router R3. Then try to open an SSH session to the router again, but this time try to log in with the username **Admin01** and a bad password. From the console window, the debug output should look similar to the following.

```
Feb  3 14:26:40.960: AAA/BIND(00000FB9): Bind i/f
Feb  3 14:26:40.960: AAA/AUTHEN/LOGIN (00000FB9): Pick method list 'SSH_LINES'
```

What message was displayed on the SSH client screen?

g. Turn off all debugging using the **undebug all** command at the privileged EXEC prompt.

Reflection

1. Why would an organization want to use a centralized authentication server rather than configuring users and passwords on each individual router?

   Centralized authentication is significantly more scalable than local database management because it minimizes the administrative time and effort required to maintain user lists across large networks

2. Contrast local authentication and local authentication with AAA.

   Standard local authentication simply verifies users against a fixed internal router database, whereas AAA local authentication provides more granular control over the login process and allows for the definition of specific fallback procedures.

Router Interface Summary Table

Router Model	Ethernet Interface #1	Ethernet Interface #2	Serial Interface #1	Serial Interface #2
1900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
2900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
4221	Gigabit Ethernet 0/0/0 (G0/0/0)	Gigabit Ethernet 0/0/1 (G0/0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)
4300	Gigabit Ethernet 0/0/0 (G0/0/0)	Gigabit Ethernet 0/0/1 (G0/0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)

Note: To find out how the router is configured, look at the interfaces to identify the type of router and how many interfaces the router has. There is no way to effectively list all the combinations of configurations for each router class. This table includes identifiers for the possible combinations of Ethernet and Serial interfaces in the device. The table does not include any other type of interface, even though a specific router may contain one. An example of this might be an ISDN BRI interface. The string in parenthesis is the legal abbreviation that can be used in Cisco IOS commands to represent the interface.